



HELWAN NATIONAL UNIVERSITY

Robotics and Mechatronics Department

Hydraulic Circuit Design and Simulation

Subject: Hydraulic & Pneumatic Control

Course Code: MEC2307

Student Name:

Student ID:

Yousef Ahmed Elbeltagy

922230111

Mohamad Sherif Shabrawy

922230084

TO: Dr. Ahmed Kadry

Introduction

This report presents the design and simulation of hydraulic control systems for three different applications. Each application represents a practical system that uses hydraulic cylinders to perform a specific task.

The required hydraulic circuits were designed using standard symbols and then implemented and tested using FluidSIM-H software to verify their operation. For each system, the circuit behavior is analyzed through simulation, and modifications are applied where required, including replacing manual valves with solenoid-operated valves and developing the corresponding control logic.

The three applications covered in this report are:

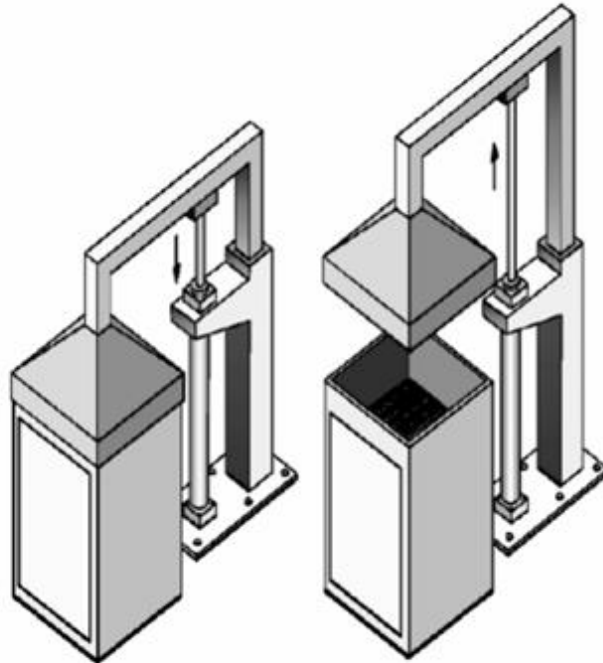
Application 1 — Hardening Furnace Cover: A single-acting cylinder raises and lowers a furnace cover, controlled by a 3/2 directional control valve with a 10 kg load representing the weight of the cover.

Application 2 — Furnace Door Control: A double-acting cylinder opens and closes a furnace door, controlled by a 4/2 directional control valve with spring return.

Application 3 — U-Shape Bending Device: A double-acting cylinder performs a bending operation on sheet metal, with two independent pushbuttons and adjustable speed control in both directions.

Application 1: Hardening Furnace Cover Control

Application Overview



This application involves the design and simulation of a hydraulic control system for a hardening furnace cover. The cover is raised and lowered by a single-acting hydraulic cylinder. A 10 kg weight is attached to the cylinder rod to simulate the actual load of the furnace cover, and this weight acts as the return force — bringing the cover back down when hydraulic pressure is released. The system is first designed using a 3/2 lever-actuated, spring-return valve, then modified in Part (b) to use a solenoid-operated valve with a corresponding electrical ladder diagram.

Part (a) — Manual Hydraulic Circuit

Introduction

In this part, the hydraulic circuit is designed using a 3/2 lever-actuated, spring-return directional control valve to control the single-acting cylinder. When the operator actuates the lever, hydraulic pressure extends the cylinder and raises the furnace cover against the weight of the load. When the lever is released, the valve returns to its normal position, pressure is relieved, and the 10 kg load pulls the cylinder rod back down by gravity, lowering the cover.

Components Used

Component	Description
Single-acting cylinder	Extends under hydraulic pressure, retracts under the weight of the 10 kg load
3/2 lever-actuated, spring-return valve	Directs pressurized flow to extend the cylinder, returns to tank connection when released
Hydraulic Power Unit (HPU)	Electric motor + hydraulic pump + internal relief valve + pressure gauge
Safety relief valve	External valve for additional overpressure protection
Load	10 kg moving mass (98.1 N) — represents the furnace cover weight and acts as the return force

Circuit Operation

Extension Stroke — Cover Raises

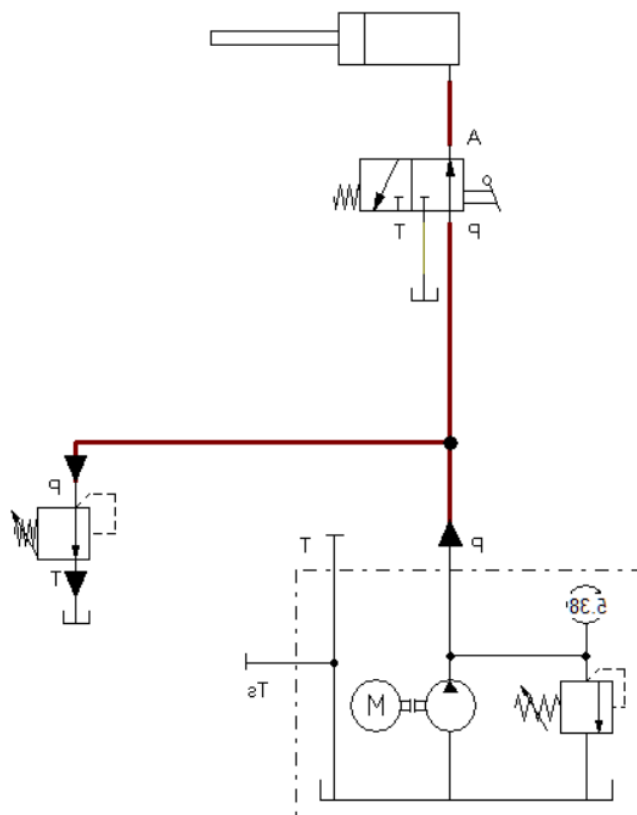


Figure 1 — FluidSIM simulation: Extension stroke (lever actuated, cover raising)

When the operator actuates the lever, the 3/2 directional control valve shifts from its normal position, connecting port P to port A. Pressurized hydraulic fluid flows from the pump through the valve and into the single-acting cylinder, pushing the piston upward against the 10 kg load and raising the furnace cover. The red flow lines in the simulation confirm that pressurized fluid is being delivered to the cylinder during this stroke.

Return Stroke — Cover Lowers

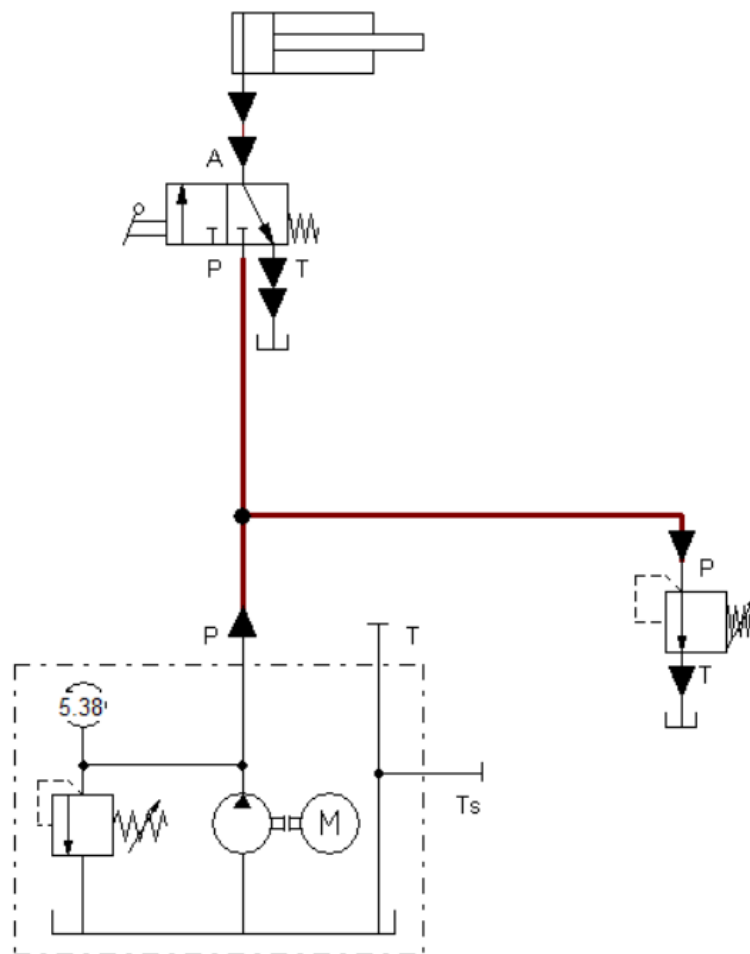


Figure 2 — FluidSIM simulation: Return stroke (lever released, cover lowering)

When the lever is released, the spring returns the valve to its normal position, connecting port A to port T. With hydraulic pressure relieved, the 10 kg weight acting on the cylinder rod pulls the piston back down by gravity, returning the hydraulic fluid to the tank. The furnace cover lowers back to its original closed position — no spring is needed for retraction.

Part (b) — Solenoid-Operated Circuit and Ladder Diagram

Introduction

In this part, the lever-actuated valve is replaced with a 3/2 solenoid-operated, spring-return valve. The solenoid (Y) is controlled electrically through a pushbutton, allowing the operator to control the furnace cover remotely without manually actuating a lever. The corresponding ladder diagram illustrates the electrical control logic of the system.

Components Used

Component	Description
Single-acting cylinder	Extends under hydraulic pressure, retracts under the 10 kg load
3/2 solenoid-operated, spring-return valve	Solenoid Y shifts the valve when energized, spring returns it when de-energized
Pushbutton	Normally open contact, closes when pressed to energize solenoid Y
Solenoid Y	Energizes to shift the valve and allow flow to the cylinder
Hydraulic Power Unit (HPU)	Electric motor + hydraulic pump + internal relief valve + pressure gauge
Safety relief valve	External valve for additional overpressure protection

Circuit Operation

Extension Stroke — Cover Raises

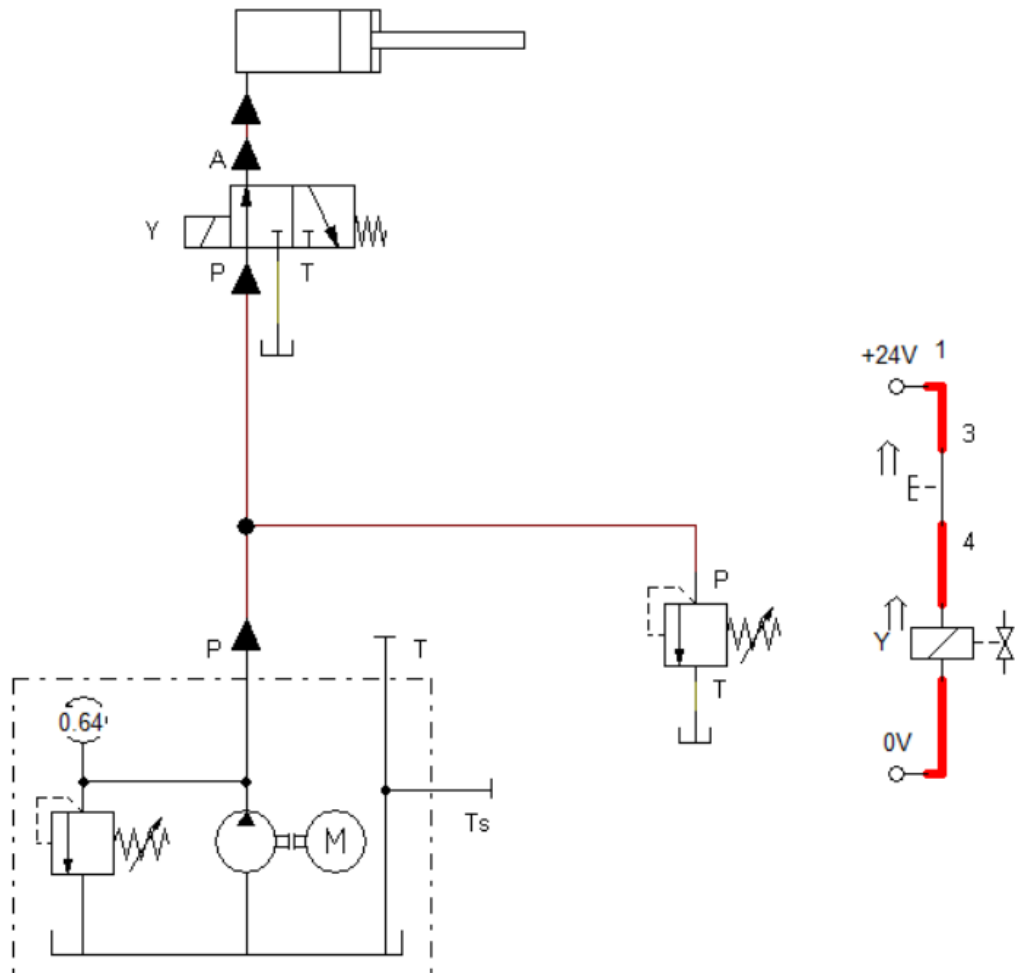


Figure 3 — FluidSIM simulation: Extension stroke (solenoid Y energized, pushbutton pressed, cover raising)

When the operator presses pushbutton, the electrical circuit is completed and solenoid Y is energized. This shifts the 3/2 valve, connecting port P to port A. Pressurized hydraulic fluid flows from the pump into the cylinder, pushing the piston upward against the 10 kg load and raising the furnace cover. The red highlighted path in the ladder diagram confirms that the circuit is active and solenoid Y is energized.

Return Stroke — Cover Lowers

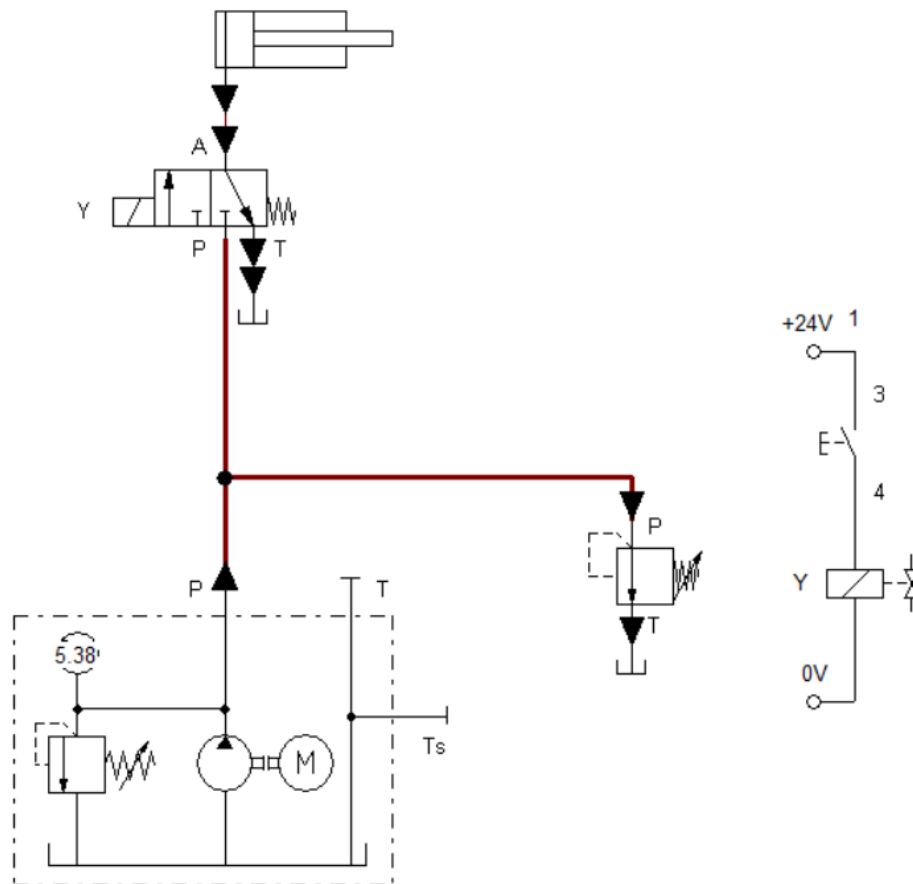


Figure 4 — FluidSIM simulation: Return stroke (solenoid Y de-energized, E released, cover lowering)

When pushbutton is released, the electrical circuit opens and solenoid Y is de-energized. The spring returns the valve to its normal position, connecting port A to port T. With hydraulic pressure relieved, the 10 kg load pulls the cylinder rod back down by gravity, returning the oil to the tank and lowering the furnace cover back to its closed position.

Load Configuration in FluidSIM

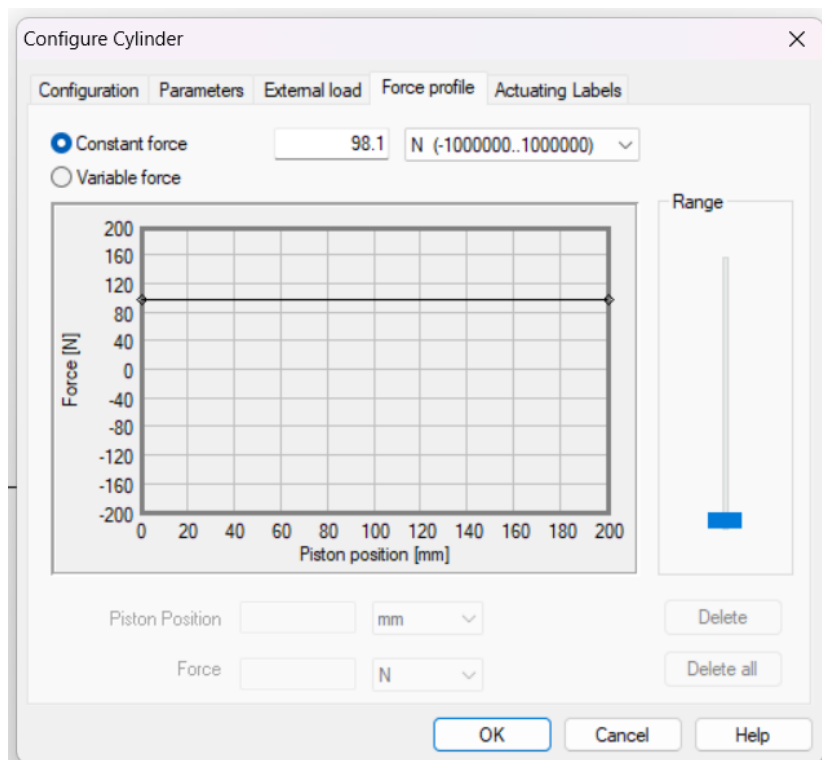


Figure 5 — FluidSIM: Constant force of 98.1 N applied to represent the 10 kg furnace cover load

The load on the cylinder was configured in FluidSIM-H using the Force Profile tab. A constant force of 98.1 N was applied to the cylinder, representing the weight of the furnace cover ($F = m \times g = 10 \times 9.81 = 98.1 \text{ N}$). This constant force acts downward on the cylinder rod at all times, opposing the extension stroke and driving the return stroke when hydraulic pressure is released — accurately simulating the real behavior of the system.

Conclusion — Application 1

In this application, a hydraulic control system was successfully designed and simulated for a hardening furnace cover using a single-acting cylinder. The 10 kg load attached to the cylinder rod represents the actual weight of the furnace cover, which also serves as the natural return force when hydraulic pressure is released.

In Part (a), the system was controlled using a 3/2 lever-actuated, spring-return valve. The circuit proved to be simple and effective — the operator manually actuates the lever to raise the cover, and releasing it allows the load to bring the cover back down.

In Part (b), the lever valve was replaced with a 3/2 solenoid-operated, spring-return valve, controlled electrically through a pushbutton and solenoid. This modification improves the system by allowing safer and more convenient remote operation. The ladder diagram confirmed the straightforward control logic — pressing the button raises the cover, releasing it lowers it.

Both circuits were successfully simulated in FluidSIM-H, verifying correct operation under the applied load conditions.

Application 2: Furnace Door Control

Application Overview

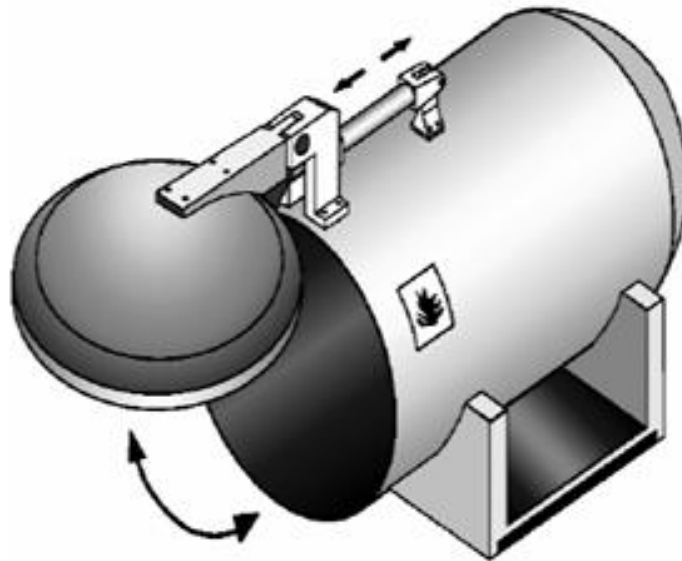


Figure 6 — Application 2: Furnace Door Control problem statement

This application involves the design and simulation of a hydraulic control system for a furnace door. The door is opened and closed by a double-acting hydraulic cylinder, controlled by a 4/2-way lever-actuated, spring-return directional control valve. The spring return ensures that the door opens only as long as the valve is actuated — when the lever is released, the spring automatically returns the valve to its normal position and the door closes again.

Part (a) — Manual Hydraulic Circuit

Introduction

In this part, the hydraulic circuit is designed using a 4/2 lever-actuated, spring-return directional control valve to control the double-acting cylinder. When the operator actuates the lever, pressurized fluid extends the cylinder and opens the furnace door. When the lever is released, the spring returns the valve to its normal position, reversing the flow and closing the door automatically.

Components Used

Component	Description
Double-acting cylinder	Extends and retracts under hydraulic pressure in both directions
4/2 lever-actuated, spring-return valve	Controls flow direction to open and close the door, spring returns to normal position when released
Hydraulic Power Unit (HPU)	Electric motor + hydraulic pump + internal relief valve + pressure gauge
Safety relief valve	External valve for additional overpressure protection

Circuit Operation

Extension Stroke — Door Opens

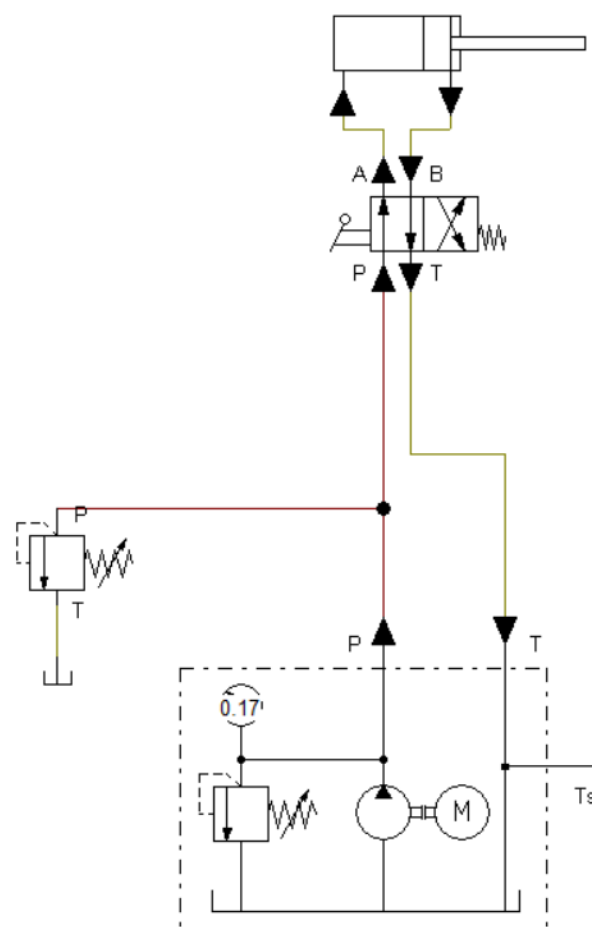


Figure 7 — FluidSIM simulation: Extension stroke (lever actuated, door opening)

When the operator actuates the lever, the 4/2 directional control valve shifts from its normal position, connecting port P to port B and port A to port T. Pressurized hydraulic fluid flows into the rod side of the cylinder, extending the piston rod and opening the furnace door. The return oil from the blank end flows back through port A to the tank. The colored flow lines in the simulation confirm the active flow path during this stroke.

Return Stroke — Door Closes

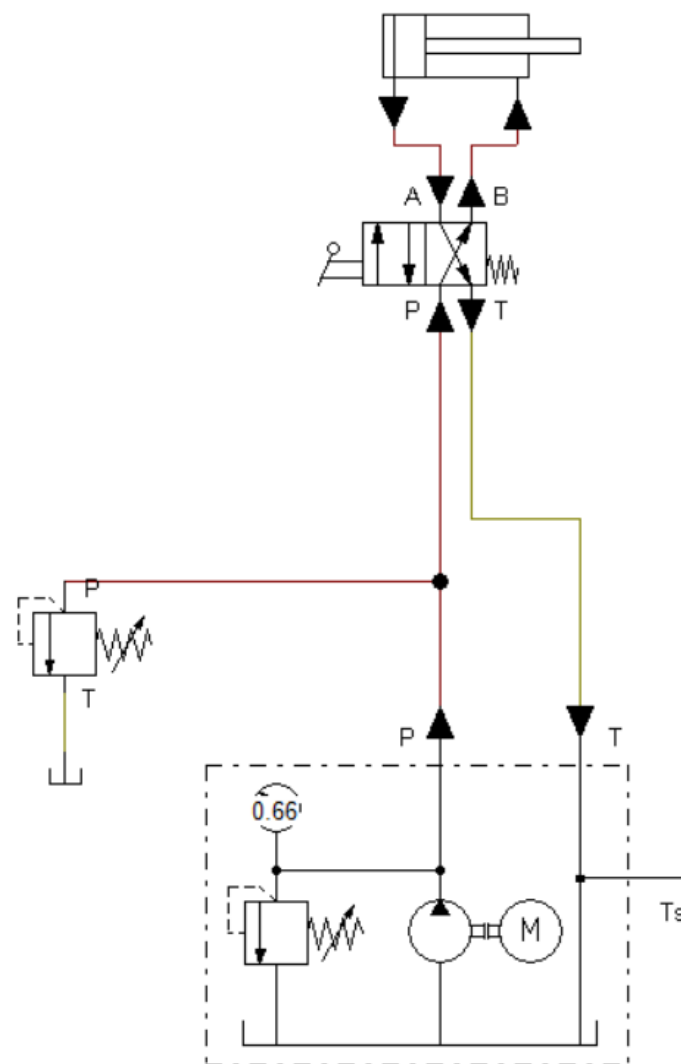


Figure 8 — FluidSIM simulation: Return stroke (lever released, door closing)

When the lever is released, the spring returns the valve to its normal position, connecting port P to port A and port B to port T. Pressurized fluid now flows into the blank end of the cylinder, retracting the piston rod and closing the furnace door. The return oil from the rod side flows back through port B to the tank. The door remains closed as long as the lever is not actuated.

Part (b) — Solenoid-Operated Circuit and Ladder Diagram

Introduction

In this part, the lever-actuated valve is replaced with a 4/2 solenoid-operated, spring-return valve. The solenoid (Y) is controlled electrically through a pushbutton, allowing the operator to open the furnace door remotely without manually actuating a lever. When the button is pressed, the solenoid energizes and the door opens. When the button is released, the spring returns the valve to its normal position and the door closes automatically.

Components Used

Component	Description
Double-acting cylinder	Extends and retracts under hydraulic pressure in both directions
4/2 solenoid-operated, spring-return valve	Solenoid Y shifts the valve when energized, spring returns it when de-energized
Pushbutton	Normally open contact, closes when pressed to energize solenoid Y
Solenoid Y	Energizes to shift the valve and allow flow to open the door
Hydraulic Power Unit (HPU)	Electric motor + hydraulic pump + internal relief valve + pressure gauge
Safety relief valve	External valve for additional overpressure protection

Circuit Operation

Extension Stroke — Door Opens

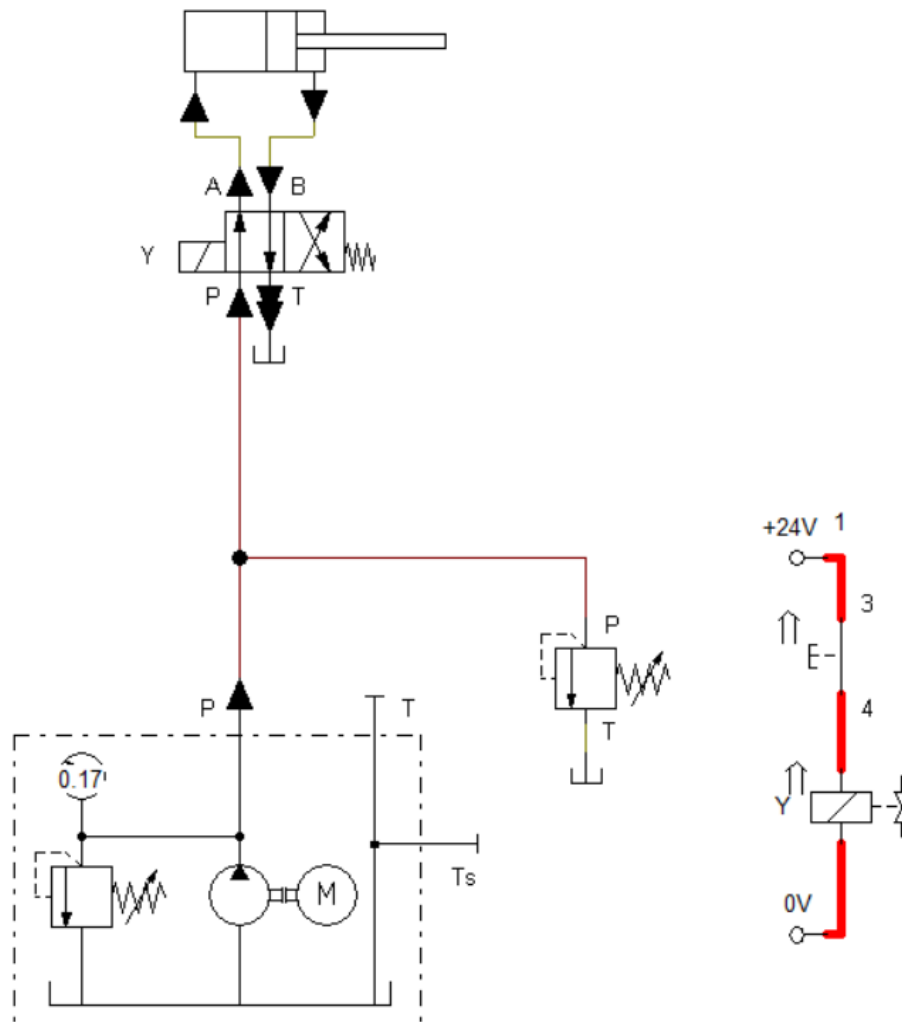


Figure 9 — FluidSIM simulation: Extension stroke (solenoid Y energized, E pressed, door opening)

When the operator presses pushbutton, the electrical circuit is completed and solenoid Y is energized. This shifts the 4/2 valve, connecting port P to port B and port A to port T. Pressurized hydraulic fluid flows into the rod side of the cylinder, extending the piston and opening the furnace door. The red highlighted path in the ladder diagram confirms that the circuit is active and solenoid Y is energized.

Return Stroke — Door Closes

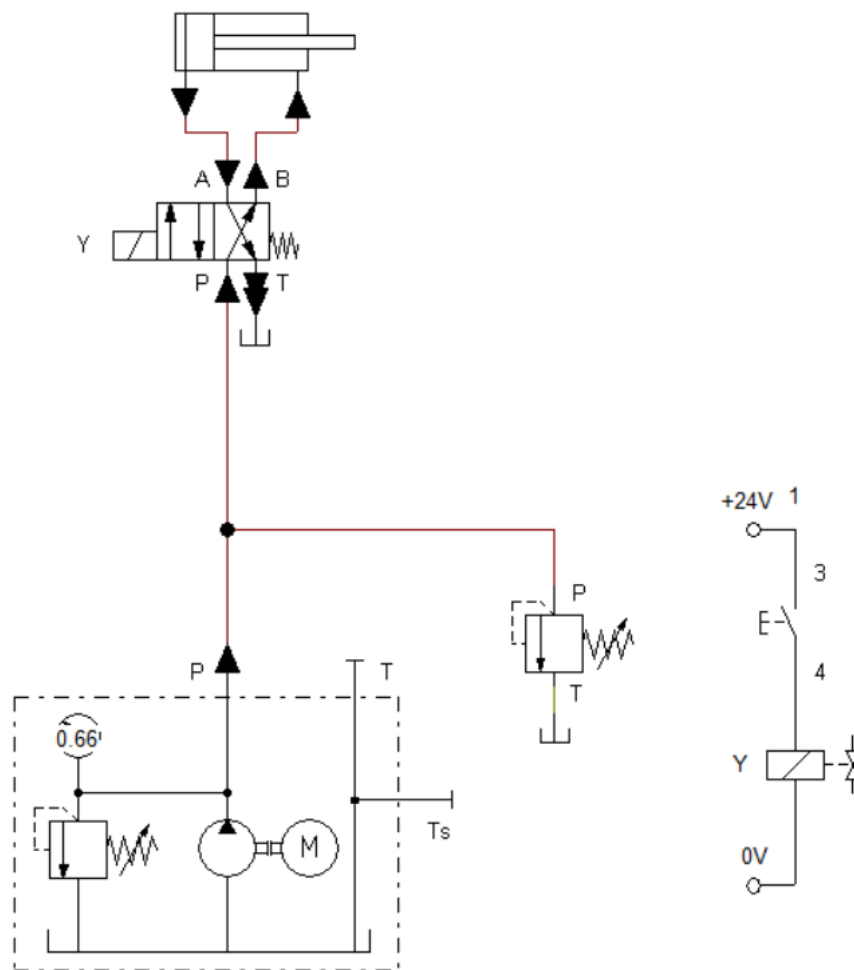


Figure 10 — FluidSIM simulation: Return stroke (solenoid Y de-energized, E released, door closing)

When pushbutton is released, the electrical circuit opens and solenoid Y is de-energized. The spring returns the valve to its normal position, connecting port P to port A and port B to port T. Pressurized fluid now flows into the blank end of the cylinder, retracting the piston and closing the furnace door automatically.

Conclusion — Application 2

In this application, a hydraulic control system was successfully designed and simulated for a furnace door using a double-acting cylinder. Unlike Application 1, the double-acting cylinder uses hydraulic pressure in both directions — to both open and close the door — making it suitable for applications where a controlled closing force is required.

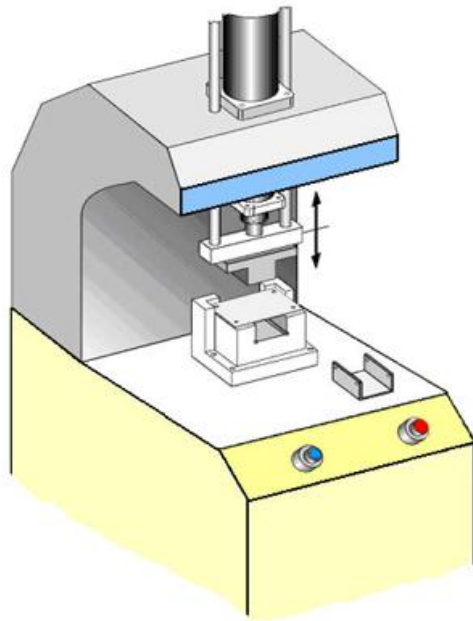
In Part (a), the system was controlled using a 4/2 lever-actuated, spring-return valve. The circuit proved effective — the operator manually actuates the lever to open the door, and releasing it allows the spring to return the valve and hydraulically close the door in a controlled manner.

In Part (b), the lever valve was replaced with a 4/2 solenoid-operated, spring-return valve, controlled electrically through a pushbutton and solenoid. This modification allows safer and more convenient remote operation. The ladder diagram confirmed the straightforward control logic — pressing the button opens the door, releasing it closes it automatically.

Both circuits were successfully simulated in FluidSIM-H, verifying correct operation and proper flow direction in both the extension and retraction strokes.

Application 3: U-Shape Bending Device

Application Overview



This application involves the design and simulation of a hydraulic control system for a bending device used to produce U-shaped sheet metal workpieces. The bending operation is performed by a double-acting hydraulic cylinder, controlled by a 4/2 solenoid valve with solenoids on both sides (X and Y) — no spring return. Two separate pushbuttons control the advance and return strokes independently. Additionally, two flow control valves with check valves are installed on both ports to regulate the speed of both strokes at a slow adjustable rate.

Hydraulic Circuit and Ladder Diagram

Introduction

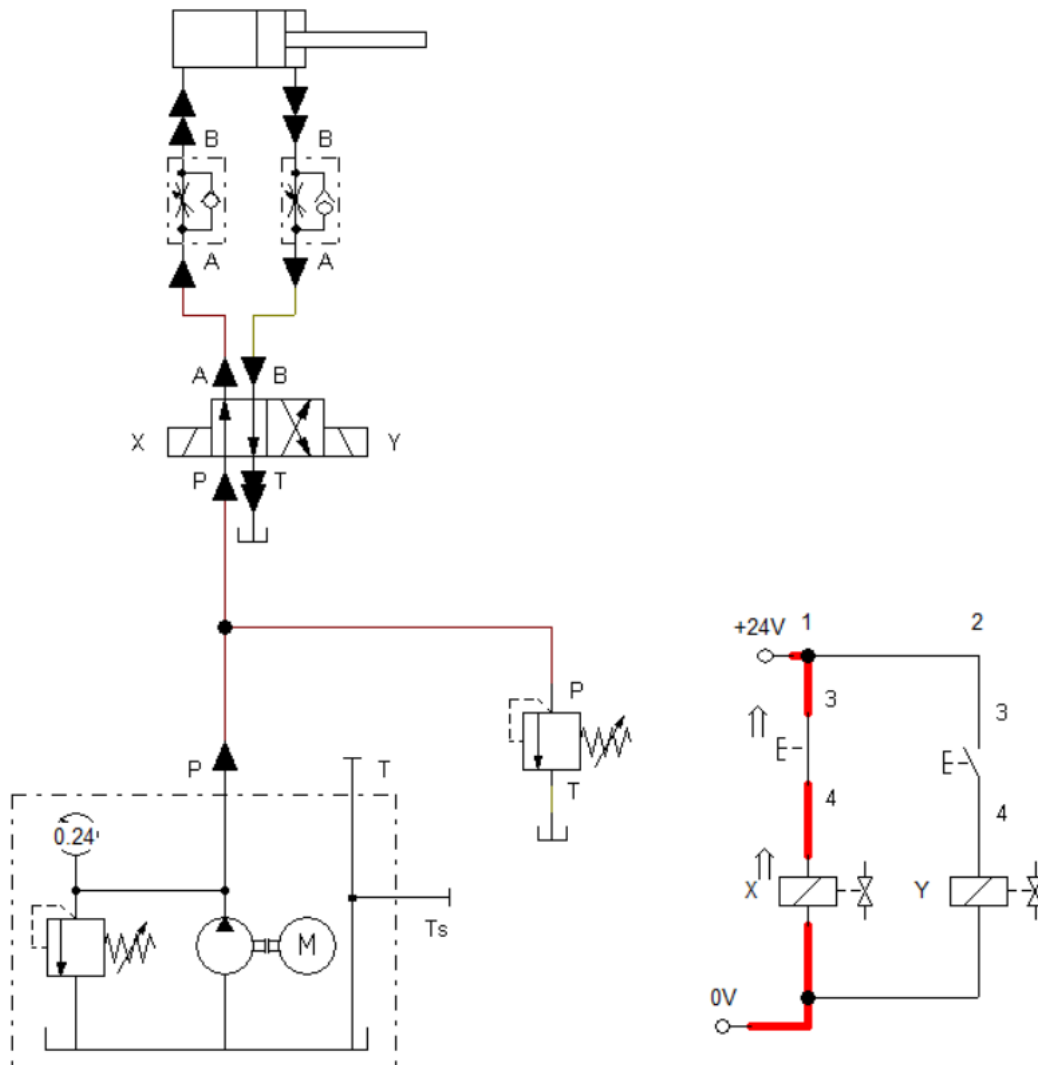
Unlike Applications 1 and 2, this application uses a 4/2 solenoid valve without spring return, meaning the valve stays in its last position until the other solenoid is energized. This is necessary because the two strokes are controlled by two independent pushbuttons — pushbutton 1 initiates the advance stroke by energizing solenoid X, and pushbutton 2 initiates the return stroke by energizing solenoid Y. Flow control valves on both ports ensure that both strokes are executed at a slow adjustable speed as required.

Components Used

Component	Description
Double-acting cylinder	Extends and retracts under hydraulic pressure in both directions
4/2 solenoid valve (X and Y)	No spring return — stays in position until opposite solenoid is energized
Pushbutton E1	Normally open contact, energizes solenoid X to advance the cylinder
Pushbutton E2	Normally open contact, energizes solenoid Y to return the cylinder
Solenoid X	Energizes to shift the valve and extend the cylinder (advance stroke)
Solenoid Y	Energizes to shift the valve and retract the cylinder (return stroke)
Flow control valve (port A)	Controls the speed of the advance stroke
Flow control valve (port B)	Controls the speed of the return stroke
Hydraulic Power Unit (HPU)	Electric motor + hydraulic pump + internal relief valve + pressure gauge
Safety relief valve	External valve for additional overpressure protection

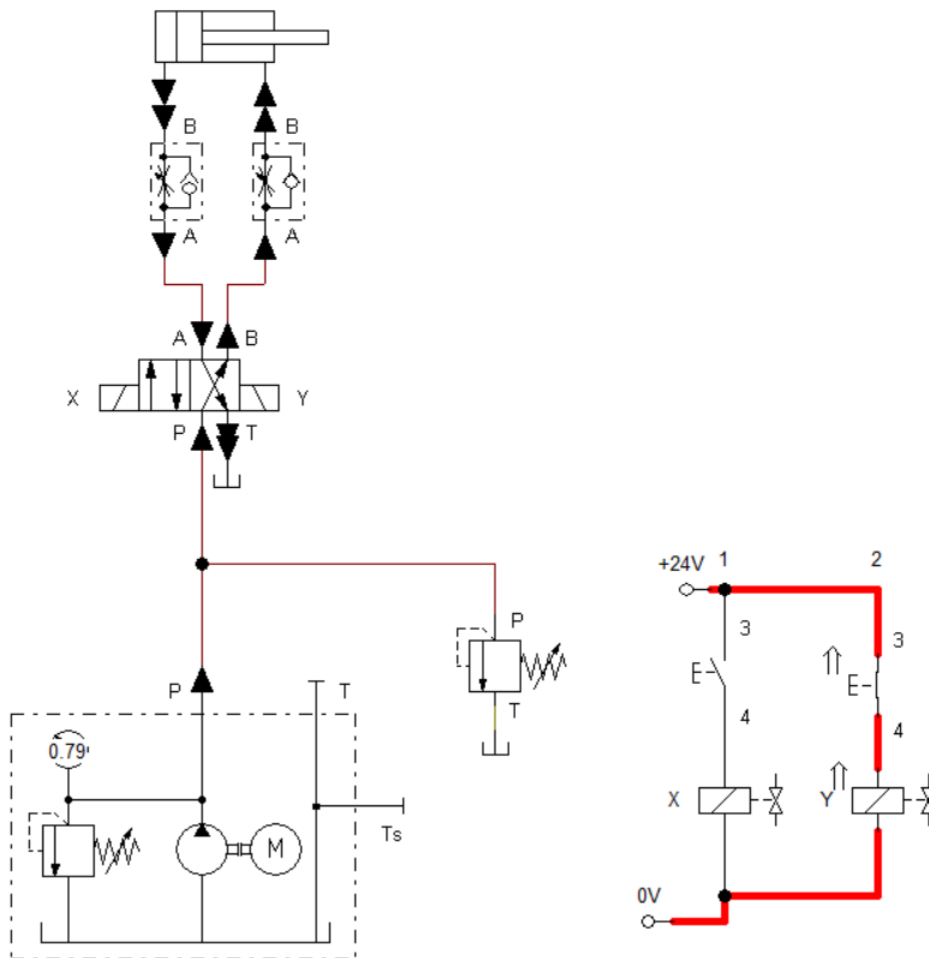
Circuit Operation

Advance Stroke — Cylinder Extends (Bending Operation)



When the operator presses pushbutton 1, solenoid X is energized, shifting the 4/2 valve to connect port P to port A and port B to port T. Pressurized hydraulic fluid flows through the flow control valve on port A at a regulated slow speed into the blank end of the cylinder, extending the piston and performing the bending operation on the sheet metal workpiece. The return oil from the rod side flows back through the flow control valve on port B to the tank. The red highlighted path in the ladder diagram confirms that rung 1 is active and solenoid X is energized.

Return Stroke — Cylinder Retracts



After the bending operation is complete, the operator presses pushbutton 2, energizing solenoid Y and shifting the 4/2 valve to connect port P to port B and port A to port T. Pressurized hydraulic fluid flows through the flow control valve on port B at a regulated slow speed into the rod side of the cylinder, retracting the piston and returning it to its original position. The return oil from the blank end flows back through the flow control valve on port A to the tank. The red highlighted path in the ladder diagram confirms that rung 2 is active and solenoid Y is energized.

Conclusion — Application 3

In this application, a hydraulic control system was successfully designed and simulated for a U-shape bending device using a double-acting cylinder. Unlike the previous applications, this system uses a 4/2 solenoid valve without spring

return, allowing the cylinder to hold its position in both directions until the operator initiates the next stroke.

The use of two independent pushbuttons — one for the advance stroke and one for the return stroke — gives the operator full control over the bending operation, ensuring the workpiece is fully shaped before the cylinder is retracted. The two flow control valves successfully regulate the speed of both strokes to a slow adjustable rate, which is essential for precise and safe bending operations.

The circuit was successfully simulated in FluidSIM-H, verifying correct operation, proper flow direction, and speed control in both the advance and return strokes.

Overall Conclusion

This report presented the design and simulation of three hydraulic control systems using FluidSIM-H software. Each application demonstrated a different aspect of hydraulic circuit design, from basic single-acting cylinder control to more complex double-acting cylinder systems with speed regulation.

Application 1 demonstrated how a single-acting cylinder can effectively control a furnace cover using gravity as the return mechanism. The simplicity of the 3/2 valve made this an ideal solution for a straightforward raise-and-lower operation. Replacing the lever with a solenoid valve in Part (b) showed how the same hydraulic logic can be implemented with electrical control for greater convenience and safety.

Application 2 showed the advantage of using a double-acting cylinder for applications requiring controlled force in both directions. The 4/2 spring-return valve ensured a fail-safe behavior — the door automatically closes when the valve is released, which is critical for furnace safety. The solenoid modification in Part (b) further improved operator safety by enabling remote control.

Application 3 introduced a more advanced control scenario, using a 4/2 solenoid valve without spring return and two independent pushbuttons for each stroke. The addition of flow control valves on both ports allowed precise speed regulation during both the advance and return strokes, which is essential for accurate sheet metal bending operations.

Overall, the simulations confirmed that all three circuits operate correctly according to their design specifications. The use of FluidSIM-H provided a reliable platform for verifying circuit behavior before physical implementation.